Bayesian Network

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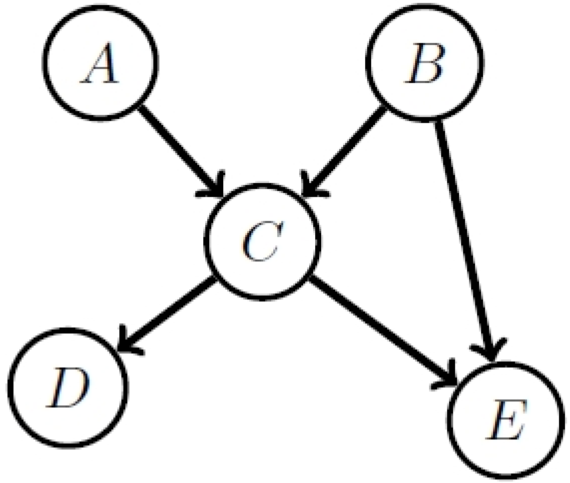
Bayesian Network

- A Bayesian network is a directed acyclic graph in which each node has associated the conditional probability of the node given its parents
- The joint distribution is obtained by taking the product of the conditional probabilities:

$$P(A, B, C, D, E) = P(A)P(B)P(C|A, B)P(D|C)P(E|B, C)$$



Example

- Sally's burglar **A**larm is sounding
- Has she been **B**urgled, or was the alarm triggered by an **E**arthquake?
- She turns the car **R**adio on for news of earthquakes.

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- Choosing an ordering:

$$
\begin{aligned}
P(A, R, E, B) &= P(A|R, E, B)P(R, E, B) \\
&= P(A|R, E, B)P(R|E, B)P(E, B) \\
&= P(A|R, E, B)P(R|E, B)P(E|B)P(B)
\end{aligned}
$$

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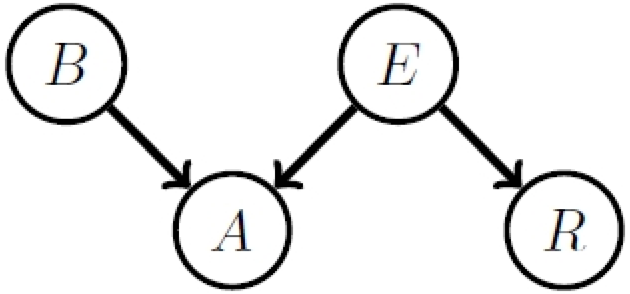
$$P(A, R, E, B) = P(A|R, E, B)P(R|E, B)P(E|B)P(B)$$

- Assumptions:
  - The alarm is not directly influenced by any report on the radio
  - The radio broadcast is not directly influenced by the burglar variable
  - Burglaries don't directly *cause* earthquakes

$$P(A, R, E, B) = P(A|E, B)P(R|E)P(E)P(B)$$

Example

$$P(A, R, E, B) = P(A|E, B)P(R|E)P(E)P(B)$$



$p(A|B, E)$

Alarm = 1	Burglar	Earthquake
0.9999	1	1
0.99	1	0
0.99	0	1
0.0001	0	0

$p(R|E)$

Radio = 1	Earthquake
1	1
0	0

Example

- Initial evidence: The alarm is sounding

$$p(B=1|A=1) = \frac{\sum_{E,R} p(B=1,E,A=1,R)}{\sum_{B,E,R} p(B,E,A=1,R)}$$
$$= \frac{\sum_{E,R} p(A=1|B=1,E)p(B=1)p(E)p(R|E)}{\sum_{B,E,R} p(A=1|B,E)p(B)p(E)p(R|E)} \approx 0.99$$

- Additional evidence: The radio broadcasts an earthquake warning

$$p(B=1|A=1,R=1) \approx 0.01$$

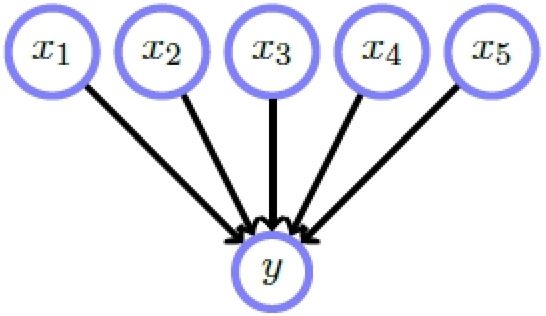
- Conclusion

  Initially, because the alarm sounds, Sally thinks that she's been burgled.
  However this probability drops dramatically when she hears that there has been an earthquake

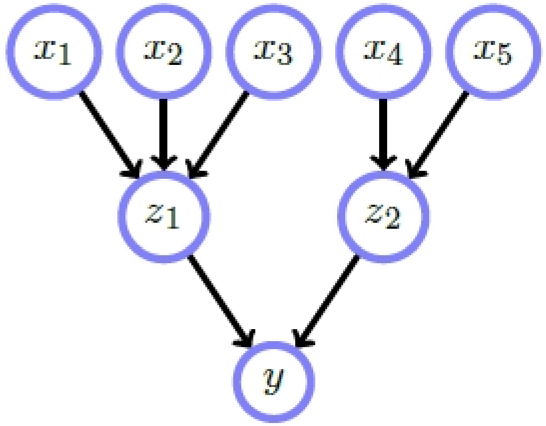
Table Size

To compute $P(y|x_1, x_2, x_3, x_4, x_5)$ where these variables are binary

- (a): $2^5 = 32$ states are required
- (b): $2^3 + 2^2 + 2^2 = 16$ states are required



(a)



(b)

Uncertain Evidence

- Soft/Uncertain evidence
  - A variable is in more than one state, with the strength of our belief about each state being given by probabilities
  - Ex) If y has the states {red, blue, green}, the vector $[0.6, 0.1, 0.3]$ could represent the probabilities of the respective states
- Hard evidence
  - A variable is in a particular state
  - Ex) All the probability mass is in one of the vector components, $[0, 0, 1]$

Uncertain Evidence

- Inference with soft-evidence can be achieved using Bayes' rule
- Writing the soft evidence as $\tilde{y}$,

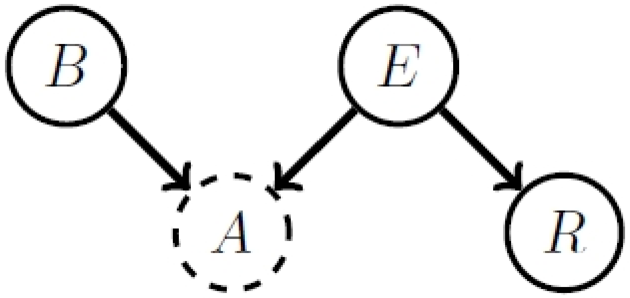
$$P(x|\tilde{y}) = \sum_{y} P(x|y)P(y|\tilde{y})$$

where $P(y = i|\tilde{y})$ represents the probability that y is in state i under the soft-evidence

Example

- Revisiting the earthquake scenario, we think we hear the burglar alarm sounding, but are not sure, specifically $P(A = tr) = 0.7$
- For this binary variable case we represent this soft-evidence for the states (tr, fa) as $\tilde{A} = (0.7, 0.3)$
- What is the probability of a burglary under this soft-evidence?

$$P\left(B = tr \middle| \tilde{A}\right) = ?$$

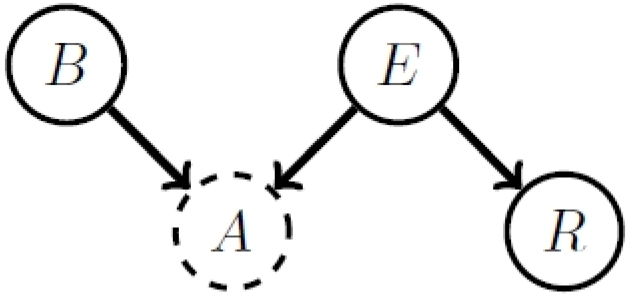


Example

- $P(A = tr) = 0.7$
- $\tilde{A} = (0.7, 0.3)$
- What is the probability of a burglary under this soft-evidence?

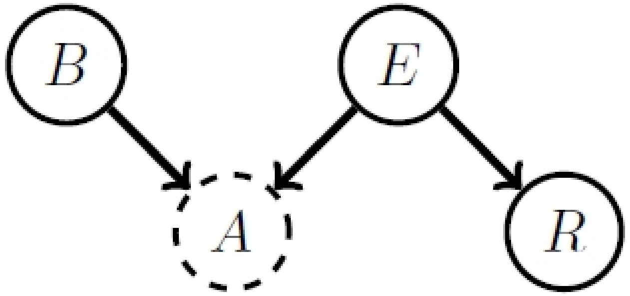
$$P\left(B = tr \middle| \tilde{A}\right) = ?$$

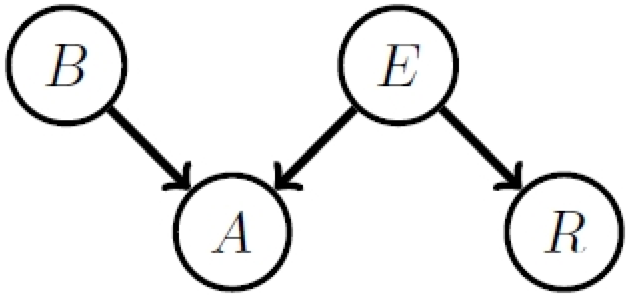
$$\begin{aligned} p(B = \mathsf{tr}|\tilde{A}) &= \sum_A p(B = \mathsf{tr}|A)p(A|\tilde{A}) \\ &= p(B = \mathsf{tr}|A = \mathsf{tr}) \times 0.7 + p(B = \mathsf{tr}|A = \mathsf{fa}) \times 0.3 \approx 0.6930 \end{aligned}$$



Example

- $P(A = tr) = 0.7$
- $\tilde{A} = (0.7, 0.3)$
- $P\left(B = tr\middle|\tilde{A}\right) = 0.6930$
- $P(B = tr|A) = 0.9900$
- Conclusion
  - This value is lower than 0.99, the probability of being burgled when we are sure we heard the alarm.
  - The probabilities $P(B = tr|A = tr)$ and $P(B = tr|A = fa)$ are calculated using Bayes' rule from the original distribution, as before.

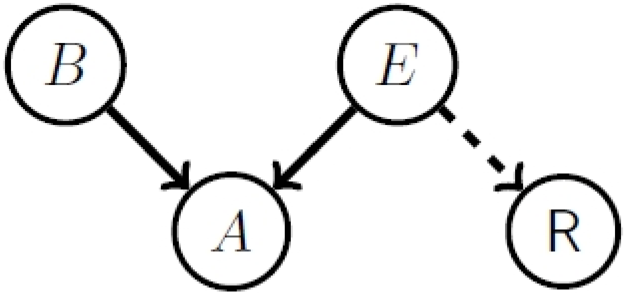




Unreliable Evidence

- Under potentially confusing reports, we decide to replace the influence of the radio variable with our own model
- We decide that we want the radio evidence to influence the inference 80% towards being an earthquake and 20% to not being an earthquake

$$p(R|E) \to p(\mathsf{R}|E) = \begin{cases} 0.8 & E = \mathsf{tr} \\ 0.2 & E = \mathsf{fa} \end{cases}$$

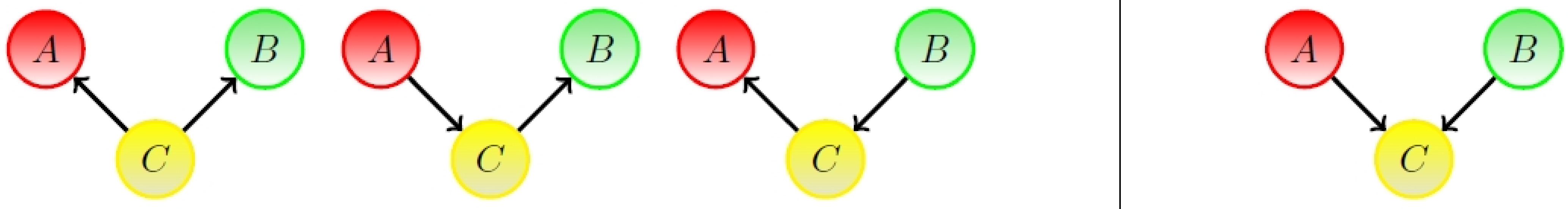


$$p(B, E, A, \mathsf{R}) = p(A|B, E)p(B)p(E)p(\mathsf{R}|E)$$

Independence in Bayesian Networks

All Bayesian networks with three nodes and two links:

$$A \perp\!\!\!\perp B \mid C \qquad\qquad A \not\perp\!\!\!\perp B \mid C$$



(a) (b) (c) (d)

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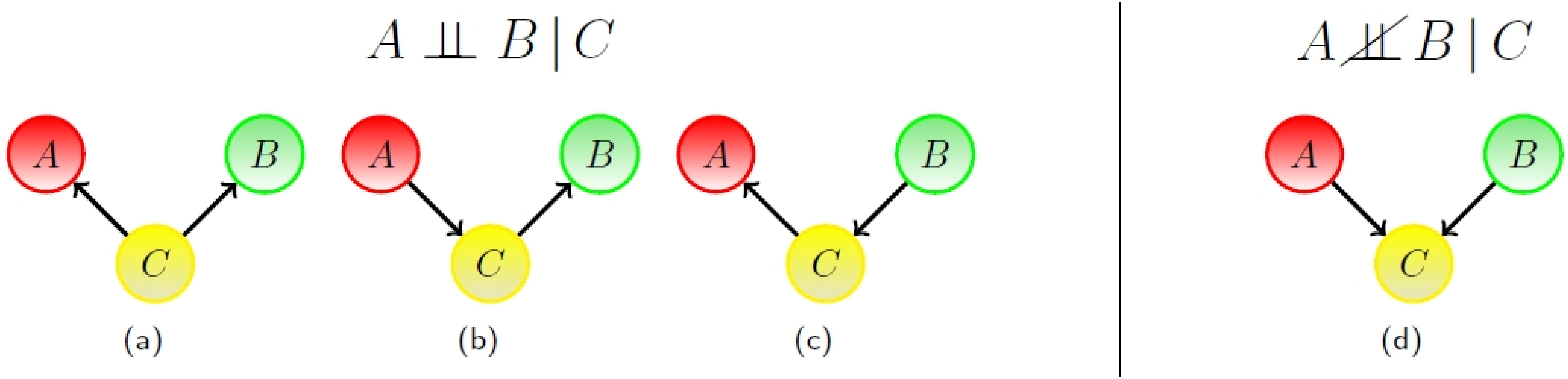

(a) $p(A, B|C) = \frac{p(A,B,C)}{p(C)} = \frac{p(A|C)p(B|C)p(C)}{p(C)} = p(A|C)p(B|C)$

(b) $p(A, B|C) = \frac{p(A)p(C|A)p(B|C)}{p(C)} = \frac{p(A,C)p(B|C)}{p(C)} = p(A|C)p(B|C)$

(c) $p(A, B|C) = \frac{p(A|C)p(C|B)p(B)}{p(C)} = \frac{p(A|C)p(B,C)}{p(C)} = p(A|C)p(B|C)$

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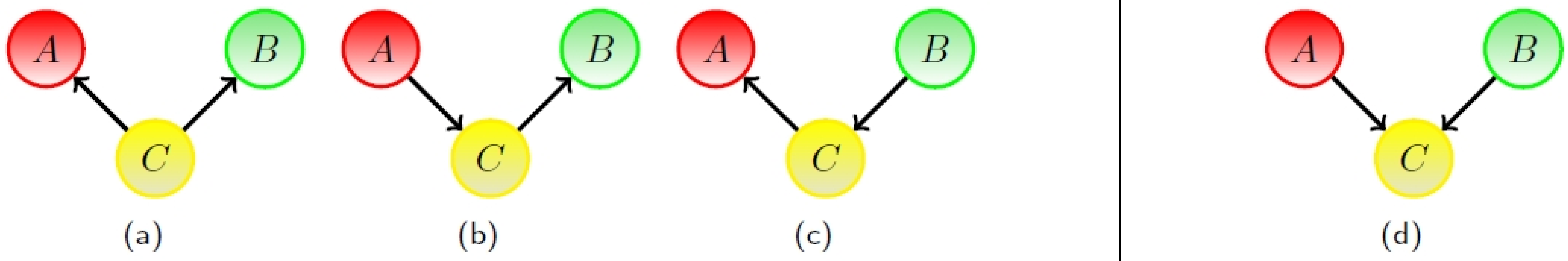
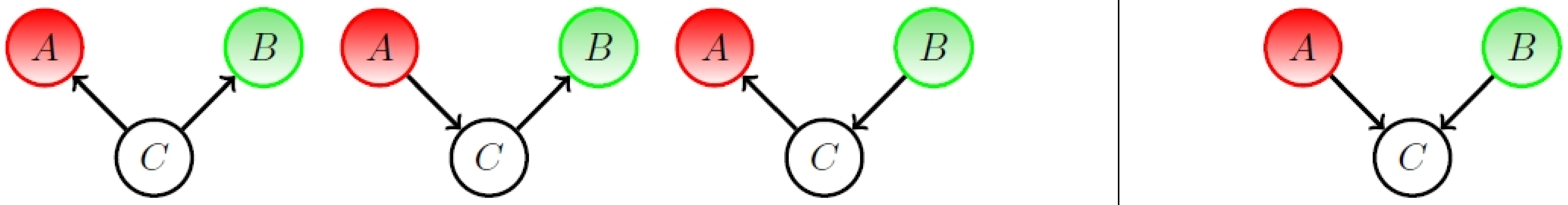


$$p(A, B|C) \propto p(C|A, B)p(A)p(B)$$

Independence in Bayesian Networks

All Bayesian networks with three nodes and two links:

$A \not\perp\!\!\!\perp B$ $\qquad\qquad$ $A \perp\!\!\!\perp B$



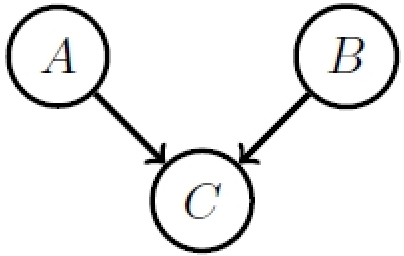
(a) (b) (c) (d)

$$p(A,B) = \sum_C p(A,B,C) = \sum_C p(A)p(B)p(C|A,B) = p(A)p(B)$$

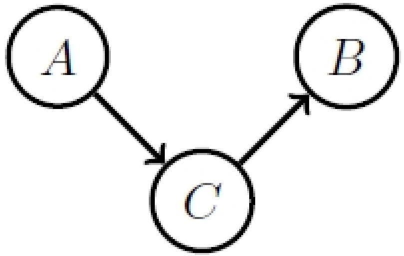
Collider

A collider contains two or more incoming arrows along a chosen path

– If C has more than one incoming link,
  then $A \perp\!\!\!\perp B$ and $A \not\!\perp\!\!\!\perp B \mid C$
  In this case, C is called collider



– If C has at most one incoming link,
  then $A \perp\!\!\!\perp B \mid C$ and $A \not\!\perp\!\!\!\perp B$
  In this case C is called non-collider



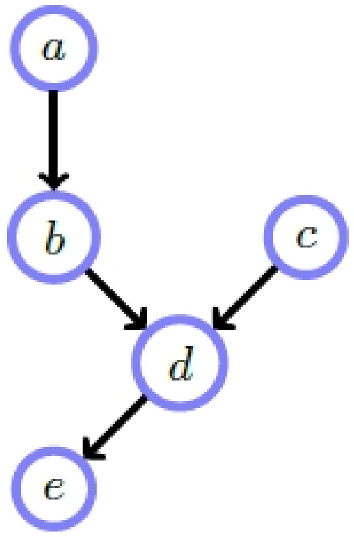
General Rule for Independence

- Given three sets of nodes (X, Y, C), if all paths from any element of X to any element of Y are blocked by C, then X and Y are conditionally independent given C
- A path P is blocked by C if at least one of the following conditions is satisfied:
  - There is a collider in the path P such that neither the collider nor any of its descendants is in the conditioning set C
  - There is a non-collider in the path P that is in the conditioning set C

Example

Is $a \perp\!\!\!\perp e \mid b$?

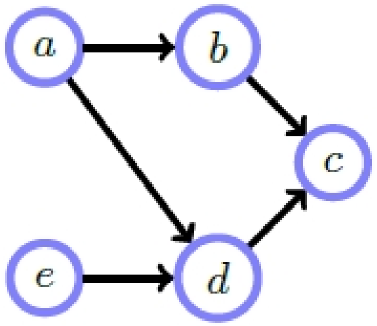
- Path a-b-d-e: *b* is non-collider and is in the conditioning set



Example

Is $a \perp\!\!\!\perp e | c$?

- Path a-d-e: *d* is a collider and is not in the conditioning set,
  a descendant of *d* is in the conditioning set
- Path a-b-c-d-e: *c* is a collider and is in the conditioning set



Causality

- Think about $P(A, B)$
  - $P(A|B)P(B)$: B causes A
  - $P(B|A)P(A)$: A causes B
- This is not very meaningful since they both represent the same distribution

Causality

- Formally BNs only make independence statements, not causal ones
- In constructing BNs, it can be helpful to think about dependencies in terms of causation since our intuitive understanding is usually framed in how one variable ‘influences’ another.

Simpson’s Paradox

- A medical trial in which patient treatment and outcome are recovered
- Two trials were conducted, one with 40 females and one with 40 males
- Question) Does the drug cause increased recovery?

Males	Recovered	Not Recovered	Rec. Rate
Given Drug	18	12	60%
Not Given Drug	7	3	70%

Females	Recovered	Not Recovered	Rec. Rate
Given Drug	2	8	20%
Not Given Drug	9	21	30%

Combined	Recovered	Not Recovered	Rec. Rate
Given Drug	20	20	50%
Not Given Drug	16	24	40%

Simpson’s Paradox

- Question) Does the drug cause increased recovery?
- Possible Answer)
  - For the males, it's best not to give the drug
  - For the females, it's also best not to give the drug
  - However, for the combined data, it's best to give the drug

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Resolving the Paradox

- We can write the distribution as

$$P(G, D, R) = P(R|G, D)P(D|G)P(G)$$

  where G: gender, D: drug, R: recovery
- Observational calculation computed $P(R|G, D)$ and $P(R|D)$
- The above formula suggests that
  - Choose a gender
  - Decide whether or not to give the drug

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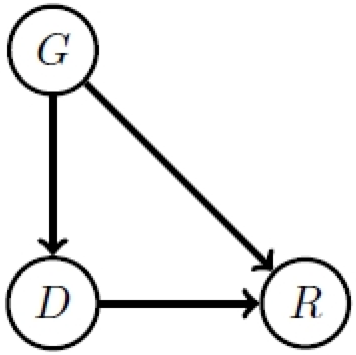


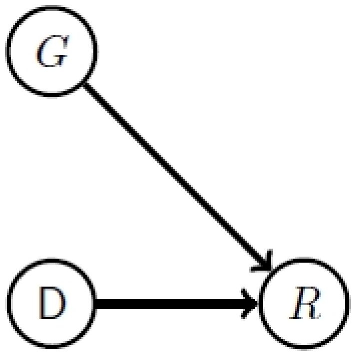
Resolving the Paradox

- We must use a distribution that is consistent with an interventional experiment

  In this case, the term $P(D|G)$ should play no role.

- We need to consider a modified distribution (conditioned on the drug)

$$\tilde{p}(G, R|D) = p(R|G, D)p(G)$$

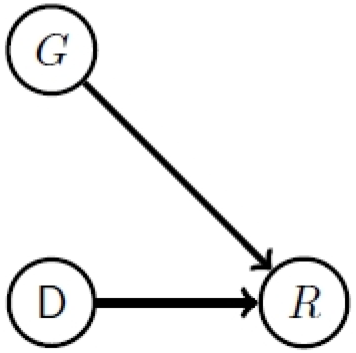


$$p(R||D) = \sum_G \tilde{p}(G, R|D) = \sum_G p(R|G, D)p(G)$$

Resolving the Paradox

$$p(\text{recovery}|\text{drug}) = 0.6 \times 0.5 + 0.2 \times 0.5 = 0.4$$
$$p(\text{recovery}|\text{no drug}) = 0.7 \times 0.5 + 0.3 \times 0.5 = 0.5$$

$$\tilde{p}(G, R|D) = p(R|G, D)p(G)$$



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Simpson’s Paradox

- The paradox occurs because we are asking a causal (interventional) question - If we give someone the drug, what happens?
- But we are performing an observational calculation
- Difference between
  - Observational evidence: given that we see
  - Interventional evidence: given that we do